FINAL REVIEW · 23CSE322 FINANCIAL ENGINEERING · TEAM 01

If I Invest *Tomorrow*

A financial decision simulator with an autonomous AI agent

Rupali K CB.AI.U4AID23033

Mahadev S CB.AI.U4AID23034

Srivatsav S CB.AI.U4AID23035

Sarang U CB.AI.U4AID23036

Indian stocks, ETFs, gold, bonds and cash · data to 1 October 2026 · demo money only

OVERVIEW

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Introduction

If I Invest Tomorrow tells an investor exactly what to buy tomorrow, how much of each asset, and the probability of reaching their goal — before they invest.

MARKET

26 NSE stocks, 3 index ETFs, gold, government bonds and a liquid fund — 12 years of daily prices.

ENGINE

Machine learning, CAPM, portfolio optimisation, 10,000-path Monte Carlo and stress tests.

AI AGENT

A local language model you talk to in plain English; it plans, simulates and invests with demo money.

> Everything is a simulation with **₹10 lakh of demo money** — no real account or broker is touched.

Objectives

1. Take the investor's **amount, horizon, risk tolerance, target and sectors**
2. **Analyse the market**: returns, risk, correlation, market regime, anomalies
3. **Estimate** expected return and risk with CAPM and machine learning
4. **Build and compare five strategies** on the efficient frontier
5. **Simulate** outcomes and **stress test** against crashes
6. Give **tomorrow's plan** in rupees and shares, and **execute** it through an AI agent

Problem statement

Given an investor's amount, horizon (1, 3 or 5 years), risk tolerance, target return and preferred sectors, decide:

1

What to buy tomorrow

2

How much of each, in rupees and shares

3

The **probability** of reaching the target, and the loss in a crash

> Constraint: monthly stock returns are close to unpredictable, so the answer must rest on **risk and probability**, not on a single price forecast.

Literature review

AREA	KEY WORK	WHAT WE USE FROM IT
Portfolio theory	Markowitz (1952); Sharpe (1964)	Efficient frontier, Sharpe ratio, CAPM expected returns
Risk estimation	Ledoit & Wolf (2004); Rockafellar & Uryasev (2000)	Stable covariance matrix; CVaR-minimising crash-resistant portfolio
Market regimes	Hamilton (1989)	Markets switch between calm and crisis states
Simulation	Glasserman (2004)	Monte Carlo for outcome probabilities
ML for returns	Gu, Kelly & Xiu (2020); Chen & Guestrin (2016)	ML adds only small, noisy return signals; tree models as baselines
Anomaly detection	Liu, Ting & Zhou (2008)	Isolation Forest for unusual market days
Robo-advisors	Beketov, Lehmann & Wittke (2018)	Automated allocation is mostly mean–variance based
LLM agents	Yao et al., ReAct (2023); Schick et al., Toolformer (2023)	Language models that call tools instead of guessing answers

Literature gap

WHAT EXISTS

- Prediction tools give a **single price forecast**, with no range of outcomes
- Robo-advisors give an allocation but **no probability of reaching a goal** and no crash test
- ML studies report **weak return predictability**, not a usable plan
- Finance chatbots **invent numbers** and cannot act safely

WHAT WE ADD

- One system: data → models → **plan in rupees and shares**
- **Probabilities**, ranges and **stress tests** for every strategy
- Honest validation: use ML only where it shows skill (volatility)
- An agent **grounded in the engine**, trading only under rules checked in code
- Built for **Indian markets**

Our models vs published research

Our numbers: walk-forward test 2021–2026 on data the models never saw · theirs: as published

TASK	PUBLISHED RESULT	OUR RESULT	VERDICT
Monthly returns: out-of-sample R^2	Best ML (neural net) R^2 **0.40%**; plain regression $-$**3.46%** — Gu, Kelly & Xiu, US	Price-pattern signal: R^2 **+0.22%**; fitted Ridge $-$0.24% (same definition)	Positive, in the range of top research
Stock direction	LSTM **54.3%** — Fischer & Krauss; daily, only the 10 most extreme picks a day, before costs	**51.5%** monthly on all 32 assets, vs 47.5% for the average	Not like for like; both near 50%
Next-month volatility	Volatility is known to be forecastable from past volatility — Corsi (HAR model)	R^2 **0.604** vs 0.50 for the best simple rule, ahead in every year	Works; target noise caps R^2 near 0.84
Outcome ranges	No common published benchmark	90% band held **84%** of 104 real outcomes (sampling range 72–94%)	Consistent with 90%

WHY NOBODY DOES MUCH BETTER

R^2 +0.22% means the forecast explains 0.22% of the month-to-month variation in returns; it is not a 0.22% return. Monthly noise ($\sim$7%) swamps the expected return ($\sim$1%).

> Our edge is not a magic forecast but **honest risk modelling**: after realistic trading costs the volatility model raises the Sharpe ratio from 0.54 to 0.73, and the full system reaches 0.80 (NIFTY 0.36).

Methodology 1 of 4: data pipeline



INVESTABLE ASSETS

32

DAILY PRICES

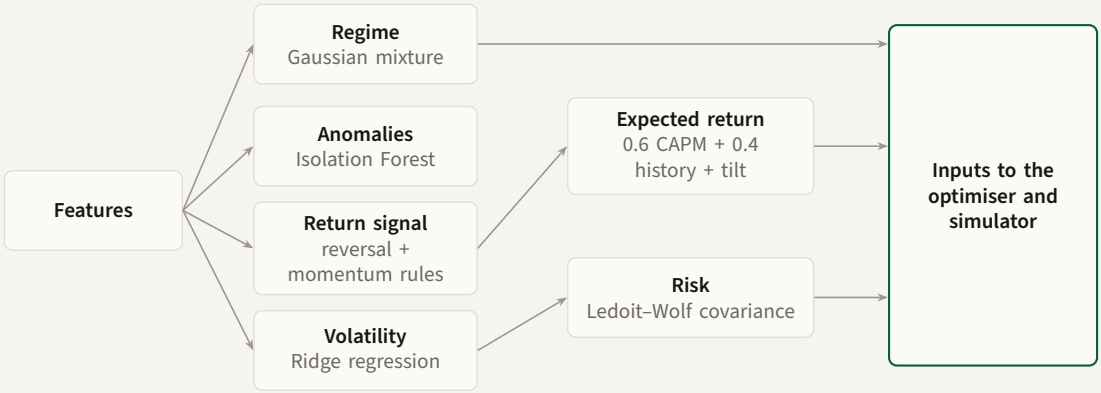
12 yrs

DATA ERRORS FIXED

3

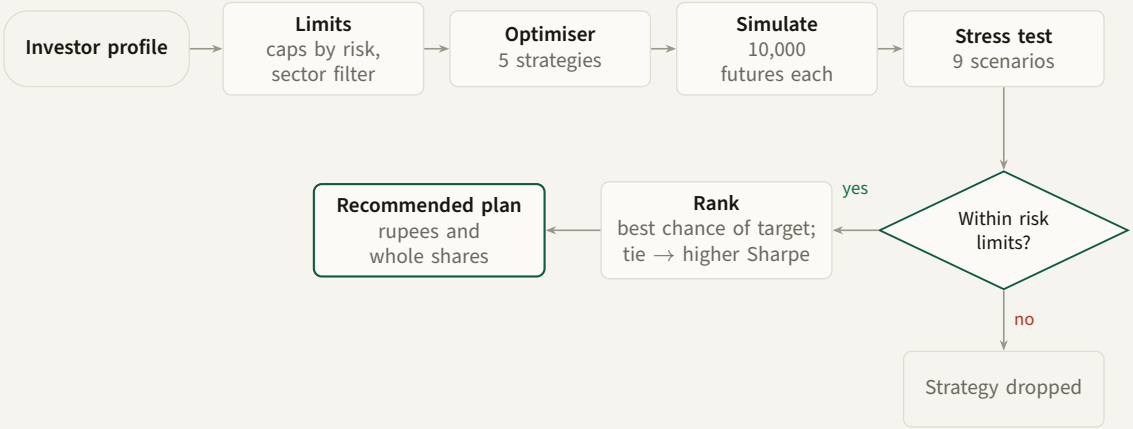
Fixed: wrong-scale prices in 3 ETFs (Dec 2019), erratic gilt-ETF prints, and a liquid-fund price that ignored daily payouts.

Methodology 2 of 4: models



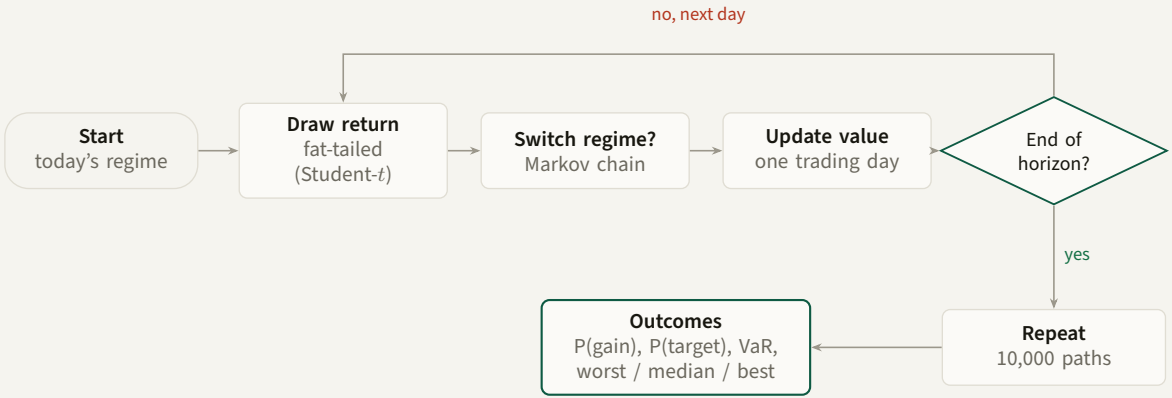
Every model is tested walk-forward on data it never saw (2021–26). Volatility is predictable (R^2 0.604, helped by each stock's results dates); for returns only a simple price-pattern signal (reversal, momentum, reversal within the sector) shows skill (IC 0.06), so it gets only a 29% weight.

Methodology 3 of 4: building and choosing the plan



Five strategies: Max Return · Min Risk · Max Sharpe · Goal-Based · Crash-Resistant (min CVaR)

Methodology 4 of 4: Monte Carlo simulation



Each path also draws its own expected return and volatility level, to include estimation error. A **stress test** starts the paths with a day-0 shock in the bear regime.

> Reality check: 84% of 104 past outcomes fell in the 90% band, within sampling error (72–94%). Misses: gold, 2020 rebound.

Working 1 of 3: four modules on one engine

Planner

Enter a profile; get tomorrow's plan, all five strategies, simulations, stress tests and a backtest.

Simulator

"What if I put ₹X, plus ₹Y a month, into these assets?" Year-by-year range vs a fixed deposit and NIFTY.

Markets

Today's regime and anomalies, risk and return of every asset, correlations, model test results.

Portfolio

₹10 lakh demo wallet: buy, sell, invest a plan, rebalance, monthly SIPs, health check.

> Example: ₹1 lakh + ₹10,000 a month for 10 years in NIFTY, gold and bonds — ₹13 lakh put in, typical result ₹19 lakh, 90% chance of profit.

Working 2 of 3: the AI agent

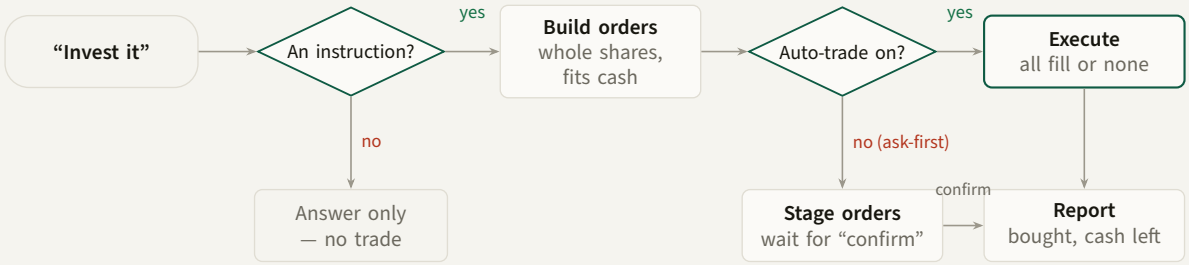
Qwen3.5-4B runs locally; it chooses the tool and explains — the engine produces every number



"I have ₹2 lakh for 3 years, medium risk, want 12%"	builds and explains tomorrow's plan
"What's my chance? Stress test it." · "Tell me about TCS"	runs the simulation, crash tests, stock analysis
"Invest it" · "Sell all my TCS" · "Check my portfolio"	trades and manages the demo wallet

23 of 23 test conversations pass. If the model skips a needed tool, the system runs it; the model only explains.

Working 3 of 3: what happens in auto-trade



AUTO-MANAGE RULES

Sell a holding down 15% · trim one that grows too big · reduce risk in a crisis · invest idle cash

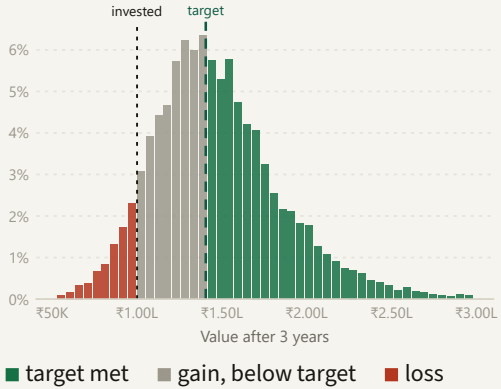
AUTOPILOT

Runs these rules automatically after every market-data update. All checks are in code, not in the language model.

Results

Sample investor: ₹1,00,000 · 3 years · medium risk · 12% target · recommended plan: Goal-Based

RANGE OF OUTCOMES AFTER 3 YEARS



BACKTEST: CHOSEN OCT 2023, HELD TO OCT 2026



EXPECTED RETURN

14.0%

BACKTEST, 3 YRS

+55%

REACH TARGET

53%

NIFTY, 3 YRS

+15%

STRESS TESTS

8 of 9

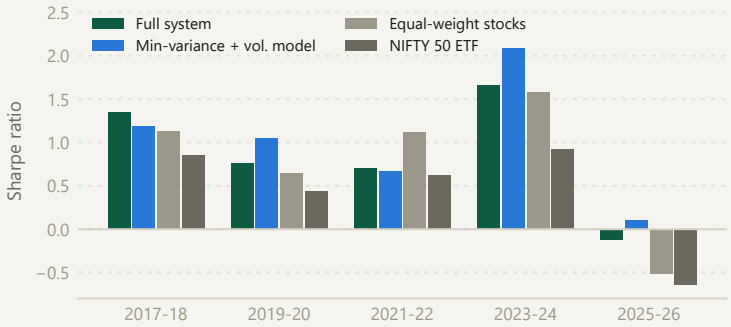
WORST FALL

−8%

Results: does it hold up?

Walk-forward 2017–2026, models refit every year · realistic Indian trading costs (STT, stamp duty, fees, spread)

SHARPE RATIO BY PERIOD, AFTER COSTS



The full system beats the NIFTY ETF in all five periods. 2025–26 runs to October 2026.

SHARPE 2021–26

0.80

EDGE OVER NIFTY

+0.43

WORST 1-YR PLAN

+2%

3-YR WINS VS NIFTY

35%

NIFTY ETF: Sharpe 0.36, worst 1-yr −21%. Edge measured 2017–26 (90% interval +0.01 to +0.83); plans from 26–34 start dates.

> The edge is **lower risk**, not higher return. Doubling costs cuts the Sharpe ratio by 0.11; starting one month later moves it by 0.1.

Results: how far can the numbers be trusted?

2017–20 chose the models · 2021–26 evaluated them, inspected repeatedly · forward test since Oct 2026

RETURN SIGNAL, AFTER COSTS

SHARPE ADDED, 2021–26

+0.07

90% interval −0.12 to +0.28; +0.04 over 2017–26. Net value **unproven**, so it keeps a small 29% weight.

MONTE CARLO BY HORIZON

HORIZON	90% BAND	INTERVAL
1 month	**89%**	85–92%
3 months	**90%**	84–95%
1 year	**85%**	75–94%

Calibrated where outcomes are many; 1 year rests on ∼7 independent windows.

WHAT MOVES P(TARGET) = 53%

Tails, regimes, vol. uncertainty	53%
Crisis correlations	49%
Expected return −2 points	44%
CAPM only / history only	28 / 80%

Sensitivity range, not a confidence interval: **44–61%** under plausible alternatives.

> Deflated Sharpe ratio (allowing for up to 300 trials): 0.81–0.89 on 2021–26 alone, 0.95–0.98 over 2017–26. Strong over ten years, suggestive on the evaluation years alone.

Conclusion

From *"what might happen"* to *"what to do tomorrow"*.

COMPLETE SYSTEM

1

Data, ML, CAPM, optimisation, simulation and stress tests in one pipeline.

STRATEGIES, 1 RULE

5

Compared on the same 10,000 futures and chosen transparently.

AGENT THAT ACTS

1

Plans, simulates, invests and manages — with safety checks in code.

> Key finding: next month's returns cannot be predicted reliably, but risk can — so the system's value is in measuring risk, comparing strategies and showing the full range of outcomes.

Future work: score the forward test as new data arrives, taxes, news sentiment, mutual funds, larger local language models.

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Thank *you*

Questions and live demo

Rupali K · Mahadev S · Srivatsav S · Sarang U

github.com/Tendool/If-I-invest-tomorrow

Simulation with demo money only — not investment advice.